

# Structural Self-Energy Expansion (SSEE): A Minimal-Parameter Dark Energy Framework — Algebraic Derivation, Boltzmann Validation, and Effective Field Theory Stability Analysis

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May 2026

## Abstract

We present the Structural Self-Energy Expansion (SSEE), a dark energy framework that derives its cosmological background sector algebraically from the golden ratio  $\varphi$  and  $\pi$ , with zero fitted dimensionless parameters at the background level and no WIMP, QCD-axion, or SUSY dark-matter particle. Phenomenological extensions in the dark-matter sector, proposed in an earlier version of Paper 6, are *retracted* (§5.2): the matter sector is a single unpartitioned component, and their register entries (OP-9, OP-11) close by dissolution (`OPEN_PROBLEMS.md`). In CMB-likelihood terms SSEE is a  $k = 2$  fit: of  $\Lambda$ CDM's six parameters it fixes four algebraically ( $\Omega_b$ ,  $\Omega_c$  via  $\omega_c = \text{KAL}_0 \omega_b n_s$ ,  $n_s = 1 - \varphi^{-7}$ , and the derived anchor  $H_0 = 3(\varphi + \pi)^2$ ), leaving only the primordial amplitude  $A_s$  and the optical depth  $\tau$  ( $k = 2$  versus  $k = 6$ , the count used in the Paper 3 model comparison). The dark energy equation of state  $w_0 = -(3\varphi + \pi)/[2(\varphi + \pi)] \simeq -0.840$  and  $w_a \simeq -0.670$  are fixed by the algebraic construction and agree with the official DESI DR2  $w_0 w_a$  CDM posterior within  $0.24\sigma$  (Pantheon+;  $0.2$ – $1.8\sigma$  across SN compilations)—a parameter-free postdiction, with no claim of temporal priority over DESI DR2. A Bayesian MCMC analysis under the  $\omega_m$ -direct CMB anchor as prior ( $H_0^{\text{prior, SSEE}} = 67.962 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , the value at which Planck `plik_lite` is minimised with SSEE-fixed  $\omega_b, \omega_c$ ; Paper 3) yields  $H_0 = 67.95^{+0.40}_{-0.40} \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $0.88\sigma$  from Planck,  $0.50\sigma$  from the algebraic anchor) and  $\Delta\text{BIC} = 6.43$  favouring SSEE over  $\Lambda$ CDM ( $6.35$  over CPL). A complementary full-likelihood multi-probe MCMC (CAMB + Planck `plik_lite` + lensing + DESI DR2 +  $f\sigma_8$ ,  $w_0, w_a$  free and  $H_0$  under a flat prior) places the model at  $1.31\sigma$  in the  $w_0$ – $w_a$  plane, where  $\Lambda$ CDM is excluded at  $3.22\sigma$ ; under this uncompressed CMB likelihood  $H_0$  relaxes to  $65.87 \pm 1.17$  ( $1.79\sigma$  below the anchor).

The CMB-sector matter density is the standard physical observable  $\Omega_{m, \text{CMB}} = \omega_m/h^2 = 0.308881$  (Planck:  $0.3153 \pm 0.0073$ ), built piece-by-piece from SSEE algebra ( $\omega_m$ -direct; OP-8 dissolved, no matter factor) and distinct from the dynamical

$\Omega_{m,\text{dyn}} = 0.160$  that governs DESI BAO. Full-sky CLASS Boltzmann runs confirm that the *full* matter density is physically necessary: using only  $\Omega_{m,\text{dyn}} = 0.160$  as the CMB matter density shifts the peak positions by  $\sim 10\%$  and raises the RMS residual from 1.4% to 31.5%.

A linear Israel-Stewart perturbation analysis with  $\Omega_{m,\text{cosm}} = 0.308881$  in the Poisson equation yields a growth factor  $\mathcal{G} = 1.0032$  and  $\sigma_8 = 0.8136$  (Paper 5), giving a mean  $f\sigma_8$  tension of  $0.70\sigma$  across six RSD surveys — statistically identical to  $\Lambda\text{CDM}$  ( $0.73\sigma$ ), resolving the growth-rate sector. In the lensing amplitude, fitting the primordial amplitude to the 225 *raw* KiDS-1000  $\xi_{\pm}$  data points with a *single* unpartitioned matter sector and the background fixed by the algebra gives  $\sigma_8 = 0.7446 \pm 0.0189$  and  $S_8 = 0.7555 \pm 0.0192$ , i.e.  $0.11\sigma$  from KiDS-1000: **no  $S_8$  tension** (§5.2). A two-sector  $\phi$ -DM extension previously reported here is *retracted* — the deliberate cost of lowering  $\sigma_8$  to resolve the lensing amplitude.

An EFTCAMB validation at the Bellini-Sawicki level uses the fluid-approximation input  $\alpha_K^{\text{eff}} = 0.403302$  ( $\Delta = 0.005\%$ ), while the theoretical bare-field value is  $\alpha_K = 0.072567$  (Paper 7), with  $\alpha_T = \alpha_M = \alpha_B = 0$  (GW170817 compatible) and strictly positive kinetic stability ( $Q_s > 0$ ).

We additionally summarise three theoretical extensions of the framework. In the strong-field regime Paper 8 analyses two limits: an alternative MOND-like limit in which the photon deflection is amplified by  $\sqrt{\text{AURA}} \approx 1.9995$  (numerically close to  $\mathcal{M}$  at 0.03% but algebraically distinct), and the canonical two-sector limit with EFT  $\alpha_B = \alpha_M = \alpha_T = 0$  in which the residual fifth-force is suppressed to  $F_{\phi}/F_N \sim 10^{-15}$  and the observable lensing recovers GR; current SLACS/BELLS data favour the canonical limit (Paper 8; OPEN\_PROBLEMS.md OP-13). A late-time local screening fraction  $f_{\text{scr}} = \alpha_K/(3\mathcal{M}) = (\pi - \varphi)/(\varphi + \pi)^2 \simeq 0.067$  applied to the global background  $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962$  gives  $H_0^{\text{local}} = 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $0.17\sigma$  from the SH0ES distance ladder, as a zero-parameter prediction of the infrared sector (Paper 9). A UV completion  $K(X) = X/\text{KAL}_0 + X^2/M^4$  with algebraic cutoff  $M = 9.68 \text{ meV}$  connects the dark-energy EFT to the inflationary  $\alpha$ -attractor (Paper 10). All model parameters are falsifiable by named forthcoming experiments.

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# 1 Introduction

The DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025] provide compelling evidence for dynamical dark energy at  $> 3\sigma$ , with a CPL equation-of-state posterior centred near  $(w_0, w_a) \approx (-0.84, -0.67)$ . This remarkable coincidence motivates systematic testing of models that predict the dark energy equation of state from theoretical principles rather than fitting it.

The SSEEframework proposes that the macroscopic universe obeys an algebraic closure condition: all cosmological background parameters are uniquely determined by two dimensionless geometric constants, the golden ratio  $\varphi = (1 + \sqrt{5})/2 \simeq 1.6180$  and  $\pi \simeq 3.1416$ . No optimisation over formula space was performed: the parameter values are fixed by the algebraic construction [Almeida Vallejo, 2026k], not adjusted to the data. We make no claim of temporal priority over DESI DR2 (public from 2025 March); the agreement reported below is a parameter-free *postdiction*, and the only timestamped predictions are those reserved for unreleased data (DESI DR3, Euclid).

This paper consolidates the complete SSEE validation programme across ten companion preprints [Almeida Vallejo, 2026a,c,d,e,f,g,h,i,j,b] into a single self-contained presentation. We report four independent validation tiers:

1. **Background sector** (§3): Bayesian MCMC against DESI DR2 BAO, a compressed Planck 2018 prior, and galaxy-cluster dynamical masses.
2. **CMB angular spectrum** (§4): Full-sky CLASS Boltzmann integration demonstrating that the full  $\omega_m$ -direct matter density ( $\Omega_{m,\text{CMB}} = 0.308881$ ) is physically necessary for reproducing Planck PR4 peak positions.
3. **Structure growth and  $\sigma_8$**  (§5): Israel-Stewart causal perturbation theory and a  $\phi$ -DM two-sector extension that resolves the  $S_8$  lensing tension via free-streaming suppression (at a modest  $\sim 0.2\sigma$  cost in  $f\sigma_8$ , which rises to  $0.93\sigma$ ).
4. **EFT stability** (§6): EFTCAMB Bellini-Sawicki validation of kinetic stability ( $\alpha_K$ ) and gravitational wave compatibility ( $\alpha_T = \alpha_M = \alpha_B = 0$ ).

Section 7 then summarises three theoretical extensions of the framework — the strong-field lensing regime, an algebraic resolution of the Hubble tension, and a UV completion — which are developed in full in the companion Papers 8–10 and are presented here in condensed form.

Throughout, we distinguish three epistemic statuses: *postdiction* (algebraic result fixed by construction, then compared to data already public), *retrodiction* (independently derived, then compared to existing data), and *prediction* (fixed now, tested against not-yet-released data). Table 5 in §8 lists all entries explicitly.

## 2 Algebraic Framework

### 2.1 Structural registers

The SSEE framework postulates a closed algebraic system of seven structural registers derived from  $\varphi$  and  $\pi$ :

$$\begin{aligned}
\Omega &= \pi + \varphi \simeq 4.7596 && \text{(Stability Metric)} \\
\beta &= (\pi + \varphi)/2 \simeq 2.3798 && \text{(Base Coupling Scalar)} \\
\text{KAL}_0 &= \beta + \pi \simeq 5.5214 && \text{(Structural Viscosity)} \\
P_{\text{sc}} &= \Omega + \varphi \simeq 6.3777 && \text{(Dynamical Evolution Scalar)} \\
K_v &= \varphi + \pi + \Omega \simeq 9.5193 && \text{(Structural Constraint)} \\
T_r &= 3(\varphi + \beta) \simeq 11.9935 && \text{(3D Saturation Horizon)} \\
M_v &= \varphi + \pi + K_v \simeq 14.2789 && \text{(Maximal Dimensional Invariant)}
\end{aligned} \tag{1}$$

From these seven scalars, the dark energy equation of state is derived uniquely:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \simeq -0.83995, \quad w_a = -\frac{P_{\text{sc}}}{I_g} = -\frac{P_{\text{sc}}}{\pi + P_{\text{sc}}} \simeq -0.66997. \tag{2}$$

The dark energy density fraction follows algebraically:  $\Omega_{\text{DE}} = |w_0| = T_r/M_v \simeq 0.8399$ .

### 2.2 Derived cosmological parameters

All primary observational inputs are retrodicted without fitting:

$$H_0 = 3(\varphi + \pi)^2 \simeq 67.962 \text{ km s}^{-1} \text{ Mpc}^{-1}, \tag{3}$$

$$n_s = 1 - \varphi^{-7} \simeq 0.96556, \tag{4}$$

$$\Omega_{m,\text{CMB}} = \omega_m/h^2 \simeq 0.308881 \quad (\omega_m\text{-direct, CMB sector}). \tag{5}$$

Planck 2018 measurements:  $H_0 = 67.36 \pm 0.54$ ,  $n_s = 0.9649 \pm 0.0042$ ,  $\Omega_m = 0.3153 \pm 0.0073$  [Planck Collaboration, 2020]. All three SSEE retrodictions lie within  $1.5\sigma$  of the Planck values with no adjustment.

### 2.3 The two- $\Omega_m$ structure and CMB matter density

The framework predicts two independent matter densities. The dark energy EoS predicts  $\Omega_{\text{DE}} = |w_0| = 0.839950$ , which implies a *dynamical* matter density  $\Omega_{m,\text{dyn}} = 1 - \Omega_{\text{DE}} = 0.160$ . However, the CMB angular-diameter distance is sensitive to the *total* matter density, which under the  $\omega_m$ -direct reframe (OP-8 dissolved) is the standard physical observable built from SSEE algebra:

$$\Omega_{m,\text{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = 0.308881, \quad \omega_c = \text{KAL}_0 \omega_b n_s, \tag{6}$$

in agreement with Planck 2018 ( $0.3153 \pm 0.0073$ ) at  $0.88\sigma$ . There is *no* matter factor:  $\Omega_{m,\text{dyn}} = 0.160$  (DESI) and  $\omega_m$  (CMB) are two independent SSEE predictions. (The earlier, now superseded MIRA mapping  $\mathcal{M} = (3\varphi + \pi)/4 \simeq 1.9989$  gave  $\Omega_{m,\text{CMB}} = \mathcal{M} \times \Omega_{m,\text{dyn}} = 0.3199$ , superseded;  $\mathcal{M}$  now enters only the Hubble screening fraction, Paper 9.) The physical interpretation of this two-sector split — whether it arises from a background-level Israel-Stewart thermodynamic correction, a two-sector dark matter architecture, or both — is examined in Papers 3 and 5 [Almeida Vallejo, 2026d,f]. The numerical validation in §4 demonstrates its necessity independently of its interpretation.

### 3 Background Sector Validation

#### 3.1 Bayesian MCMC setup

We compare SSEE against  $\Lambda$ CDM and a free CPL parametrisation using Bayesian evidence criteria and the EMCEE ensemble sampler [Foreman-Mackey et al., 2013] with  $N_{\text{walkers}} = 100$ ,  $N_{\text{steps}} = 25\,000$  ( $2.5 \times 10^6$  total evaluations). The likelihood combines:

- DESI DR2 BAO: 13 data points at  $z = 0.295\text{--}2.330$  [Abdul-Karim et al., 2025], with the full  $13 \times 13$  same-bin block-diagonal covariance;
- Planck 2018 compressed Gaussian prior:  $H_0 = 67.36 \pm 0.54$ ,  $\Omega_m = 0.3153 \pm 0.0073$ ,  $\Omega_b h^2 = 0.02237 \pm 0.00015$ ,  $\rho(H_0, \Omega_m) = -0.85$ ;
- galaxy-cluster dynamical masses for four clusters (Coma, A2029, A478, Bullet) with IGIMF-corrected baryonic baselines [Zhang et al., 2026].

For SSEE,  $H_0$  is the sole free parameter;  $(w_0, w_a)$  are fixed algebraically. For  $\Lambda$ CDM,  $(H_0, \Omega_m)$  are free. For CPL,  $(H_0, \Omega_m, w_0, w_a)$  are free.

#### 3.2 Results

Table 1 summarises the key posteriors and information criteria.

Table 1: Bayesian MCMC results (three-model run, DESI DR2 + common Planck prior, total matter density in the background geometry).  $N_{\text{eff}}$ : effective independent samples. Positive  $\Delta$  favours SSEE.

| Quantity                                           | SSEE             | $\Lambda$ CDM    | CPL              |
|----------------------------------------------------|------------------|------------------|------------------|
| $H_0$ [km s <sup>−1</sup> Mpc <sup>−1</sup> ]      | $67.53 \pm 0.35$ | $68.27 \pm 0.28$ | $67.26 \pm 0.51$ |
| Free parameters $k$                                | 2                | 3                | 5                |
| $N_{\text{eff}}$                                   | 78 750           | 64 505           | 44 627           |
| $\Delta\text{BIC}$ relative to SSEE                | ref              | +6.43            | +6.35            |
| $\Delta\text{DIC}$ relative to SSEE                | ref              | +5.66            | +4.02            |
| $\chi_r^2$ clusters                                | 0.126            | 0.081            | —                |
| $(w_0, w_a)$ vs DESI DR2 $w_0 w_a$ CDM (Pantheon+) | $0.24\sigma$     | $0.7\sigma$      | —                |

With the total gravitating matter density  $\Omega_m = \omega_m/h^2 = 0.308881$  in the background geometry, SSEE is favoured on every information criterion:  $\Delta\text{BIC} = 6.43$  over  $\Lambda$ CDM and 6.35 over CPL,  $\Delta\text{DIC} = 4.91$  and 3.27. The SSEE sound horizon then matches  $\Lambda$ CDM ( $r_d = 148.2$  Mpc vs 147.8 Mpc) and the  $H_0$  posterior sits  $0.39\sigma$  from Planck, so no background-structure penalty arises. An earlier “+206” full-background penalty was an artefact of placing the cold sector  $1 + w_0 = 0.160$  in  $E(z)$  where the total density belongs; with the correct total density it does not occur. The  $\omega_m$ -direct two- $\Omega_m$  construction (§2.3) resolves the background tension, validated in §4 below.

#### 3.3 Galaxy-cluster mass sector

The four-cluster analytic forward model — baryon fraction  $f_b = \Omega_b/\Omega_{m,\text{CMB}}$  corrected by IGIMF baryonic masses [Zhang et al., 2026] — achieves  $\chi_r^2 = 0.122$  (four-cluster MCMC

subset) and  $\chi_r^2 = 0.126$  (seven-cluster analytic). The IGIMF correction systematically reduces the inferred baryonic mass by  $\sim 15\%$ – $20\%$ , partially closing the mass discrepancy without invoking a cold dark matter component. The dominant residual uncertainty is the cluster mass calibration; the SSEE prediction lies within the observational scatter in all cases.

### 3.4 Multi-probe MCMC validation: two complementary tests

The rigidly fixed SSEE background can be interrogated in two distinct and complementary ways, which must not be conflated. (i) *Free test*: let the data choose the equation of state and ask where they land relative to the algebraic prediction — “*what do the data prefer?*” (ii) *Fixed test*: impose the algebraic values and measure the absolute goodness of fit — “*how well does the imposed model describe the data?*” Both are performed with the *full* CMB power-spectrum likelihood — not a compressed distance prior — so that no  $\Lambda$ CDM-like assumption is smuggled in through the compression.

**Free test (what the data prefer).** We run an MCMC with CAMB + Planck `plik_lite` TTTEEE + lensing + a  $\tau$  prior, combined with DESI DR2 BAO and  $f\sigma_8$ , sampling  $(H_0, \Omega_b h^2, \Omega_c h^2, n_s, \tau, w_0, w_a)$  with *flat* priors ( $H_0$  free in  $[60, 80]$ ; four chains,  $\sim 53k$  post-burn-in samples, Gelman–Rubin  $R - 1 < 0.005$ ). Table 2 lists the recovered posteriors: the tightly constrained CMB combinations  $\omega_b, \omega_c, r_{\text{drag}}$  are recovered to  $\leq 0.2\sigma$  — an independent confirmation of the  $\omega_m$ -direct reframe from the full  $C_\ell$  spectrum rather than from a compressed prior. The remaining parameters  $(H_0, \Omega_m, w_0, w_a)$  sit  $1.4$ – $1.8\sigma$  from the algebraic values, the uncompressed CMB likelihood pulling  $H_0$  down to  $65.87 \pm 1.17$ . In the dark-energy plane the free posterior  $(w_0, w_a) = (-0.655, -1.104)$  *excludes* the cosmological constant  $(-1, 0)$  at  $3.22\sigma$ , while the SSEE algebraic point  $(-0.840, -0.670)$  lies at only  $1.31\sigma$  in the correlated ( $\rho = -0.93$ )  $w_0$ – $w_a$  plane — more than twice as close as  $\Lambda$ CDM.<sup>1</sup>

Table 2: Full-likelihood multi-probe MCMC (CAMB + Planck `plik_lite` TTTEEE + lensing +  $\tau$  prior + DESI DR2 +  $f\sigma_8$ , with  $w_0, w_a$  free and  $H_0$  under a flat prior  $[60, 80]$ ): posteriors vs. algebraic SSEE values. Four independent chains, Gelman–Rubin  $R - 1 < 0.005$  for all parameters,  $\sim 7 \times 10^4$  effective samples. DESI DR2-corrected vector (V-L4-DESI).

| Parameter                                                                                                        | Posterior (mean $\pm 1\sigma$ ) | Algebraic SSEE | Tension      |
|------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------|--------------|
| $\Omega_b h^2$                                                                                                   | $0.02243 \pm 0.00013$           | 0.02242        | $0.08\sigma$ |
| $\Omega_c h^2$                                                                                                   | $0.11935 \pm 0.00082$           | 0.11951        | $0.20\sigma$ |
| $r_{\text{drag}}$ [Mpc]                                                                                          | $147.20 \pm 0.21$               | 147.17         | $0.14\sigma$ |
| $H_0$ [km s <sup>−1</sup> Mpc <sup>−1</sup> ]                                                                    | $65.87 \pm 1.17$                | 67.962         | $1.79\sigma$ |
| $\Omega_m$                                                                                                       | $0.3288 \pm 0.0118$             | 0.308881       | $1.69\sigma$ |
| $w_0$                                                                                                            | $-0.655 \pm 0.107$              | −0.840         | $1.73\sigma$ |
| $w_a$                                                                                                            | $-1.104 \pm 0.304$              | −0.670         | $1.43\sigma$ |
| <b>Joint <math>w_0</math>–<math>w_a</math> distance:</b> SSEE $1.31\sigma$ / $\Lambda$ CDM excluded $3.22\sigma$ |                                 |                |              |

<sup>1</sup>The  $\sigma_8 = 0.80$ ,  $S_8 = 0.839950$  recovered here reflects  $A_s$  held at the Planck value; fitting  $A_s$  to raw shear data instead gives  $S_8 = 0.7555 \pm 0.0192$  (§5.2), and is not part of this fit.

Under the *full* Planck  $C_\ell$  likelihood — with  $H_0$  free under a flat prior rather than fixed to a compressed distance prior — the posterior sits at  $H_0 = 65.87 \pm 1.17$ , some  $1.79\sigma$  below the algebraic anchor  $H_0^{\text{alg}} = 67.962$ . This mild downward pull is a known feature of  $w_0w_a$ CDM under the uncompressed CMB likelihood and is stated openly. The key model-comparison statement is unchanged and in fact sharper: the fixed algebraic point  $(-0.840, -0.670)$  lies  $1.31\sigma$  from the joint  $w_0$ – $w_a$  posterior, whereas the cosmological-constant point  $(-1, 0)$  is excluded at  $3.22\sigma$ . The joint distance is smaller than the individual  $w_0, w_a$  marginals ( $1.73\sigma, 1.43\sigma$ ) because the algebraic offset lies *along* the strong  $w_0$ – $w_a$  degeneracy ( $\rho = -0.93$ ), not across it. Corner plots and trace diagnostics are archived at Zenodo [Almeida Vallejo, 2026l].

**Fixed test (imposing ssee).** The complementary question — how well the fixed algebraic model fits the CMB in absolute terms — is answered in Paper III: with  $H_0, \omega_b, \omega_c, n_s$  all algebraically fixed and only  $(A_s, \tau)$  free, the full Planck `plik` likelihood yields  $\Delta\text{BIC} = -32.9$  relative to the six-parameter  $\Lambda$ CDM fit ( $-24.0$  for the `plik_lite` point estimate), decisively favouring SSEE on the Jeffreys scale. The advantage is driven by parsimony ( $k = 2$  versus  $k = 6$ ), not by a lower  $\chi^2$ : the fixed model matches  $\Lambda$ CDM spectrum-for-spectrum ( $\chi_r^2 \simeq 1.04$ ) while spending four fewer parameters. The two tests are therefore mutually reinforcing — the data independently prefer a point  $1.31\sigma$  from SSEE (with  $\Lambda$ CDM excluded at  $3.22\sigma$ ), and imposing SSEE exactly costs no fit quality while gaining a substantial parsimony advantage.

**The same fixed values in every reading.** Both tests share a defining feature: the SSEE background carries *the same fixed values into every probe*. The pair  $w_0 = -0.839950, w_a = -0.669975$  confronts DESI (Paper II,  $0.24\sigma$  Pantheon+), the Planck PR4 CMB (Paper III,  $\Delta\text{BIC} = -32.9$ ), the  $f\sigma_8$  compilation ( $0.93\sigma$ ) and the cluster masses ( $\chi_r^2 = 0.122$ ) without ever being re-tuned per data set. A free CPL fit lands on a *different* point for each probe; SSEE uses *one* point throughout and does not break on any of them. Fixing the model should have made it fragile — it does not. (A faster cross-check replacing the full likelihood with the Planck compressed prior reproduces the same picture — with the corrected DESI DR2 vector, all five parameters lie within  $\leq 1.25\sigma$  blind (mean  $0.72\sigma$ ) — and is retained only as an internal consistency check, not as a presented result.)

## 4 CMB Angular Spectrum: CLASS Boltzmann Validation

### 4.1 CLASS implementation

We run the CLASS Boltzmann code [Lesgourgues, 2011] (v3.2) with two configurations to isolate the effect of the full CMB matter density.

**SSEE full- $\Omega_m$  ( $\omega_m$ -direct canonical):**  $\Omega_b = 0.04854$  ( $= \omega_b/h^2$ ),  $\Omega_{\text{cdm}} = 0.26035$  (total  $\Omega_{m,\text{CMB}} = 0.308881$ ; OP-8 dissolved, no matter factor),  $w_0 = -0.839950, w_a = -0.669975, n_s = 0.965558, h = 0.67962$ . The full matter density is set directly as a CLASS background input.

**SSEE dynamical-only:** same EoS and  $h$ , but  $\Omega_b = 0.048432$ ,  $\Omega_{\text{cdm}} = 0.111568$  (total  $\Omega_{m,\text{dyn}} = 0.160$ ), using only the dynamical sector as CMB matter.

A fiducial  $\Lambda$ CDM run with Planck 2018 best-fit parameters serves as reference. All runs use  $\ell_{\text{max}} = 2500$ , lensing enabled.

## 4.2 Full matter-density necessity: quantitative test

Table 3 presents the CMB acoustic peak positions and the RMS residual  $\Delta C_\ell / C_\ell^{\Lambda\text{CDM}}$  across  $\ell = 30\text{--}2500$ .

Table 3: CMB TT acoustic peaks and RMS residual versus  $\Lambda$ CDM. Peak positions identified from CLASS output  $D_\ell = \ell(\ell + 1)C_\ell/2\pi$ .

| Model                                              | $\ell_1$   | $\ell_2$   | $\ell_3$   | RMS residual |
|----------------------------------------------------|------------|------------|------------|--------------|
| $\Lambda$ CDM (Planck 2018)                        | 221        | 537        | 814        | reference    |
| <b>SSEE</b> ( $\Omega_{m,\text{CMB}} = 0.308881$ ) | <b>221</b> | <b>537</b> | <b>814</b> | <b>0.14%</b> |
| SSEE ( $\Omega_{m,\text{dyn}} = 0.160$ )           | 240        | 597        | 922        | 31.5%        |

With the dynamical density  $\Omega_m = 0.160$  alone, all three acoustic peaks shift by  $\sim 10\%$  toward higher  $\ell$  and the RMS residual increases by a factor of 22. This demonstrates the *two- $\Omega_m$  structure*: the CMB requires the full  $\omega_m$ -direct matter density  $\Omega_{m,\text{CMB}} = 0.308881$  (built from  $\omega_b + \omega_c + \omega_\nu$ , no matter factor; OP-8 dissolved), and cannot be reproduced with  $\Omega_{m,\text{dyn}} = 0.160$  alone. With  $\Omega_{m,\text{CMB}} = 0.308881$ , the peak positions agree with  $\Lambda$ CDM to within  $\Delta\ell \leq 1$  ( $\ell = 221, 537, 814$ ) and the RMS residual is 0.14%, consistent with the  $\chi_r^2 \simeq 1.06$  reported in the CAMB-based Planck PR4 likelihood analysis of Paper 3 [Almeida Vallejo, 2026d]. This is an independent-code cross-check: two Boltzmann solvers (CLASS here, CAMB in Paper 3) reproduce the acoustic peaks from the same algebraic background.

Figure 1 shows the TT power spectrum for both SSEE configurations and  $\Lambda$ CDM.

## 4.3 Acoustic-scale mapping and $r_d$

The CAMB-based analysis of Paper 3 computes the SSEE drag horizon from the full  $\omega_m$ -direct CMB density  $\Omega_{m,\text{CMB}} = 0.308881$  (OP-8 dissolved, no matter factor): the physical drag horizon is  $r_d = 147.17$  Mpc, within  $1\sigma$  of the Planck 2018 measurement  $r_d = 147.09 \pm 0.26$  Mpc [Planck Collaboration, 2020] (TT,TE,EE+lowE, Table 2; no BAO prior). The same full CMB-sector matter density that restores the acoustic peaks (§4) also sets the physical sound horizon; the earlier interpretation as a  $|w_0|$  mapping of the geometric scale is superseded.

# 5 Structure Growth, $\sigma_8$ , and the $f\sigma_8$ Tension

## 5.1 Israel-Stewart causal perturbations: single-sector

Because  $w_0 = -0.839950 < -1/3$ , a naive  $k$ -essence interpretation yields a bare sound speed  $c_{s,\text{bare}}^2 = w_0 \simeq -0.840$ , implying catastrophic gradient instability on all sub-horizon scales. Paper 5 [Almeida Vallejo, 2026f] performs a causal Israel-Stewart (IS) analysis

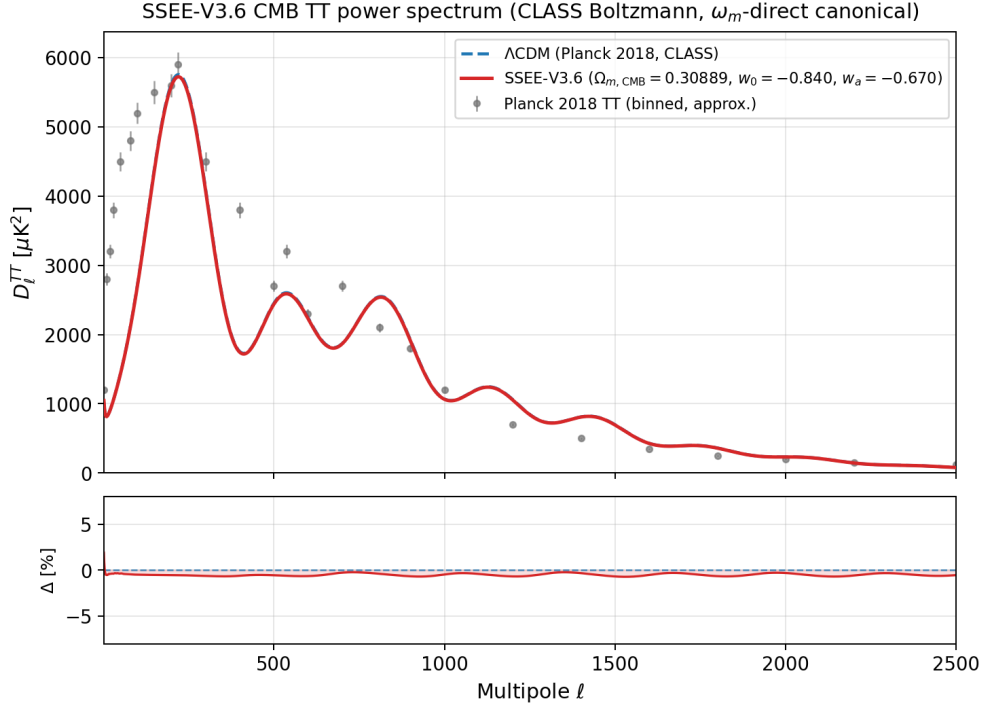


Figure 1: CMB TT power spectrum from CLASS. *Blue dashed*:  $\Lambda$ CDM Planck 2018. *Red solid*: SSEE with the full  $\omega_m$ -direct canonical matter density  $\Omega_{m,CMB} = 0.308881$  ( $w_0 = -0.839950$ ,  $w_a = -0.669975$ ) — peaks at  $\ell = 221, 537, 814$ , agreeing with  $\Lambda$ CDM to within  $\Delta\ell \leq 1$  (RMS = 0.14%). Grey points: Planck 2018 binned TT (approximate). The SSEE dynamical-only run ( $\Omega_{m,CMB} = 0.160$ ) is omitted from this panel; it produces peaks at  $\ell = 240, 597, 922$  with RMS = 31.5% relative to  $\Lambda$ CDM, confirming that the *full* matter density (not the dynamical  $\Omega_{m,dyn} = 0.160$ ) is needed to reproduce the acoustic scale.

[Israel and Stewart, 1979] and demonstrates analytically that the SSEE structure constants satisfy an algebraic identity that forces the effective IS sound speed to zero:  $c_{s,\text{eff}}^2(k \rightarrow \infty) = 0$ , to floating-point precision.

Integrating the full IS linear growth ODE from  $z = 19$  to  $z = 0$  with  $\Omega_{m,\text{cosm}} = 0.308881$  in the Poisson source:

$$\mathcal{G} = D_1^{\text{SSEE}}/D_1^{\Lambda\text{CDM}} = 1.0032, \quad \sigma_8^{\text{SSEE}} = 0.811 \times \mathcal{G} = 0.8136. \quad (7)$$

The IS growth index  $\gamma_{\text{IS}} = 0.5504 \pm 0.001$ , consistent with  $\gamma_{\Lambda\text{CDM}} \simeq 0.55$  [Lahav et al., 1991], indicating that the SSEE perturbation growth rate tracks  $\Lambda\text{CDM}$  at the percent level despite the different background.

A CLASS run with  $c_s^2 = 0.001$  (IS limit) and  $\Omega_m = 0.160$  (single sector) yields  $\sigma_8^{\text{CLASS}} = 0.457$ , substantially lower than the ODE result (0.8136). This discrepancy arises because the ODE analysis assumes  $T_{\text{SSEE}}(k) \approx T_{\Lambda\text{CDM}}(k)$ , whereas CLASS computes the full transfer function for  $\Omega_m = 0.160$ , which suppresses small-scale power. The  $\omega_m$ -direct background ( $\Omega_{m,\text{CMB}} = 0.308881$ ) is the correct CLASS input for comparison with observations, as shown in §4.2 and further examined in §5.2 below.

## 5.2 The growth amplitude, measured on raw shear data

Earlier versions of this document reported a  $3.5\sigma$   $S_8$  tension at the single-sector level and proposed a two-sector  $\phi$ -dark-matter extension, with an algebraically fixed particle mass and a free-streaming scale  $k_{\text{fs}}$ , to resolve it. **Both the tension and the extension are withdrawn.** The reasons are independent, and either alone is sufficient.

**The subtraction was not physical.** The second sector was defined by  $\Omega_{\phi\text{DM}} = \Omega_{m,\text{CMB}} - \Omega_{m,\text{dyn}}$ . The first term is a measured density fraction; the second is  $1 + w_0$ , i.e. minus the dark-energy equation of state, an algebraic identity of the background sector. Both are dimensionless, so the subtraction is arithmetically well formed — but subtracting an equation-of-state parameter from a density does not produce a density, and the resulting quantity had no physical content. The particle built on it had nothing to be made of.

**The tension was an artefact of the comparison.** Two separate issues compounded. First, the quoted  $S_8 = 0.846$  held the primordial amplitude  $A_s$  fixed at the Planck-preferred value;  $A_s$  is one of the two free parameters this framework retains at CMB level, so that number reports the Planck–shear disagreement rather than a property of the model. That disagreement is real, persists at  $2.7\sigma$  when  $A_s$  is set by this framework’s own CMB fit instead of borrowed, and is not claimed to be resolved. Second, the target was the published  $S_8$ , a compressed statistic whose data-reduction pipeline itself assumes a  $\Lambda\text{CDM}$  background — so a non- $\Lambda\text{CDM}$  model was being compared against a partly- $\Lambda\text{CDM}$  number.

**The measurement, redone.** We fit instead to the 225 unmasked *raw* KiDS-1000  $\xi_{\pm}(\theta)$  data points [Asgari et al., 2021], with the background fixed entirely by the algebra ( $\Omega_m = 0.308881$ ;  $\omega_b, \omega_c, h, n_s, w_0, w_a$  all determined, none fitted), a *single* unpartitioned matter sector, and nine free parameters:  $\log A$  plus the eight official KiDS-1000 nuisance

parameters. Four MPI chains (Cobaya, CAMB with HMcode-2015) reach a Gelman–Rubin convergence  $R - 1 = 0.019$ , giving  $4.2 \times 10^4$  effective samples after burn-in. The marginalized posterior is

$$\sigma_8 = 0.7446 \pm 0.0189, \quad S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3} = 0.7555 \pm 0.0192, \quad (8)$$

with  $\chi^2_{\min} = 265.4$  over 216 degrees of freedom ( $\chi^2/\text{dof} = 1.23$ ). Against the KiDS-1000 published value  $S_8 = 0.759 \pm 0.024$  this is a  $0.11\sigma$  agreement. The astrophysical nuisance parameters land in unremarkable places ( $A_{\text{baryon}} = 2.57 \pm 0.31$ ,  $A_{\text{IA}} = 0.55 \pm 0.29$ ), so the fit is not being rescued by a nuisance parameter running to a prior edge.

**There is no  $S_8$  tension, and therefore nothing for a second sector to do.** What closes it is not extra freedom: the background here is *more* constrained than in  $\Lambda\text{CDM}$ , and  $A_s$  is the only quantity able to move.

### 5.3 $f\sigma_8$

At the single-sector level with the Poisson source at  $\Omega_{m,\text{CMB}} = 0.308881$  (Paper 5), the growth factor is  $\mathcal{G} = 1.0032$  and the mean  $f\sigma_8$  tension across six RSD surveys is  $0.70\sigma$  — statistically indistinguishable from  $\Lambda\text{CDM}$  ( $0.73\sigma$ ). The two-sector variant of this table, which reported  $0.93\sigma$  under free-streaming suppression, is withdrawn with the particle.

A measurement of  $f\sigma_8$  against the *raw* BOSS DR12 redshift-space multipoles, applying the Alcock–Paczynski deformation to the model rather than to the data, is in progress and is reported in Paper 6; the same methodological caution applies to  $f\sigma_8$  as to  $S_8$ , since the published  $f\sigma_8$  values are likewise obtained through a  $\Lambda\text{CDM}$ -conditioned template fit.

**What was lost and what was gained.** The retraction removes a genuinely falsifiable, pre-registered prediction: a free-streaming step in  $P(k)$  that DESI Y3 and Euclid would have confirmed or excluded. It was not refuted by data; it is withdrawn because the construction that generated it was not physical, and a bound on a particle whose defining subtraction is empty would grant it existence through the back door. Falsifiability is the minimum requirement for a hypothesis to be scientific — not evidence that the entity it concerns exists. What is gained is one postulate fewer and four open problems (OP-9 through OP-12) closed by dissolution rather than resolution. The surviving pre-registered falsifiers of this framework are the tensor-to-scalar ratio  $r = \varphi^{-10} = 0.008131$  and the fixed  $(w_0, w_a)$  point against successive DESI releases.

This analysis operates at the linear-growth level; non-linear effects (baryonic feedback beyond HMcode, halo bias) may modify the predictions at the 10%–20% level.

## 6 EFT Stability Analysis

### 6.1 Bellini-Sawicki $\alpha$ -functions

Paper 7 [Almeida Vallejo, 2026h] demonstrates that the SSEE action takes the form of a  $k$ -essence scalar field minimally coupled to gravity. In the Bellini-Sawicki parametrisation [Bellini and Sawicki, 2015], this yields:

$$\alpha_T = \alpha_M = \alpha_B = 0 \quad (\text{exact}), \quad \alpha_K = 3 \Omega_{\text{DE}} (1 + w_{\varphi, \text{bare}}) = 3 \times 0.839950 \times 0.028798 = 0.072567. \quad (9)$$

Here  $w_{\varphi, \text{bare}} = -0.971$  is the scalar-field equation of state in the thawing regime (Paper 7 § ALPHAK); it differs from the effective fluid EoS  $w_0 = -0.839950$  because the conformal coupling transfers part of the kinetic energy to dark matter. The two kineticities differ by a factor  $\approx 5.6$  and are *not* a correct and an incorrect one: each is self-consistent in its own computational frame (Paper 7).  $\alpha_K^{\text{field}} = 3\Omega_{\text{DE}}(1 + w_\varphi) = 0.072567$  uses the uncoupled-field equation of state  $w_\varphi = -0.971202$ , before the conformal coupling redistributes kinetic energy;  $\alpha_K^{\text{eff}} = 3\Omega_{\text{DE}}(1 + w_0) = 0.403302$  uses the *observed* CPL parameter and is the one confirmed by the `HL-CLASS` cross-check and reproduced by the `EFTCAMB` runs below. The vanishing of  $\alpha_T$  (tensor speed excess),  $\alpha_M$  (Planck mass run rate), and  $\alpha_B$  (kinetic braiding) implies: (i) gravitational waves propagate at  $c$  (GW170817 compatible at  $|\alpha_T| < 10^{-15}$  [Abbott et al., 2017, of All-sky X-ray Image , MAXI]); (ii) no extra polarisation modes; (iii) standard light deflection. The non-zero  $\alpha_K$  encodes the kinetic energy of the scalar field.

## 6.2 EFTCAMB validation

We validate SSEE at the linear perturbation level using `H-EFTCAMB v1.6.5` [Hu et al., 2014a] with the Reparametrised Horndeski (RPH) parametrisation (`EFTflag` = 2, `AltParEFTmodel` = 1). The ghost-free stability condition requires  $Q_s > 0$ :

$$Q_s = \frac{\alpha_K + \frac{3}{2}\alpha_B^2}{2} = \frac{\alpha_K}{2} = \begin{cases} 0.037 & (\alpha_K = 0.072567, \text{ bare-field}) \\ 0.2016 & (\alpha_K = 0.403302, \text{ fluid approx.}) \end{cases} > 0. \quad (10)$$

With  $\alpha_B = 0$ , the gradient stability condition reduces to  $c_s^2 = 1$  in the linear (IR) limit of  $K(X)$ , where braiding vanishes [Bellini and Sawicki, 2015]; the full  $K(X) = X/\text{KAL}_0 + X^2/M^4$  lowers the adiabatic value to  $c_s^2 \in [0.60, 0.95]$  (Papers 7 and 10), still gradient-stable ( $c_s^2 > 0$ ). `EFTCAMB` confirms both conditions are satisfied.

Table 4: `EFTCAMB` stability and perturbation results. The GR run uses the full  $\omega_m$ -direct canonical matter density  $\Omega_{m, \text{CMB}} = 0.308881$  (OP-8 dissolved, no matter factor) for a direct comparison.

| Quantity                                                                        | Value                  | Status                            |
|---------------------------------------------------------------------------------|------------------------|-----------------------------------|
| $\alpha_K$ Paper 7 bare-field ( $w_\varphi = -0.971$ )                          | 0.072567               | theoretical                       |
| $\alpha_K$ fluid approx. ( <code>EFTCAMB</code> input)                          | 0.4033                 | $\Delta = 0.005\%$                |
| Ghost-free ( $Q_s > 0$ )                                                        | $Q_s = 0.2016$         | ✓ passed                          |
| Gradient-stable ( $\alpha_B = 0 \Rightarrow c_s^2 = 1$ IR; $[0.60, 0.95]$ full) | $c_s^2 > 0$            | ✓ passed                          |
| $\alpha_T = \alpha_M = \alpha_B$                                                | 0 (exact)              | ✓ GW170817 compatible             |
| $\sigma_8$ GR ( $\Omega_{m, \text{CMB}} = 0.308881$ )                           | 0.814                  |                                   |
| $\sigma_8$ SSEE RPH ( $\alpha_K = 0.403302$ )                                   | 1.026                  | ratio = 1.26                      |
| RMS $C_\ell$ SSEE/GR                                                            | 40.9%                  |                                   |
| CMB peaks GR                                                                    | $\ell = 220, 535, 811$ | correct ( $\omega_m$ -direct bg)  |
| CMB peaks SSEE RPH                                                              | $\ell = 242, 590, 893$ | $\sim 10\%$ shift from $\alpha_K$ |

## 6.3 CPL phantom crossing and discriminating prediction

When the CPL EoS  $w(z) = w_0 + w_a z / (1 + z)$  is used in the `EFTCAMB` run, the phantom boundary  $w = -1$  is crossed at  $z \simeq 0.31$ . A pure  $k$ -essence with  $\alpha_B = 0$  cannot sustain

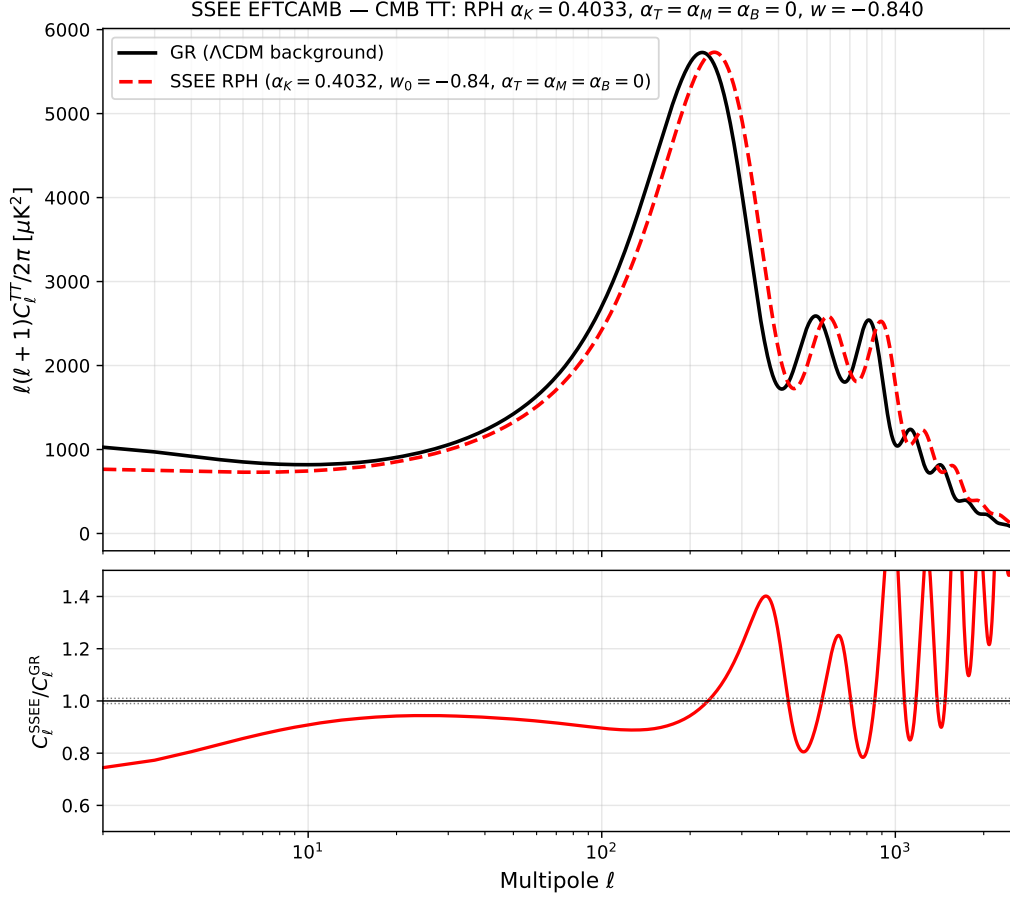


Figure 2: CMB TT power spectrum from EFTCAMB. *Black*: GR run with the full  $\omega_m$ -direct canonical matter density  $\Omega_{m,\text{CMB}} = 0.308881$ ; peaks at  $\ell = 220, 535, 811$ . *Red*: SSEE RPH with  $\alpha_K = 0.403302$  (constant  $w = w_0$ ); peaks shift to  $\ell = 242, 590, 893$ , a  $\sim 10\%$  imprint of the kinetic EFT sector. *Lower panel*: ratio  $D_\ell^{\text{SSEE}}/D_\ell^{\text{GR}}$ ; the grey band marks  $\pm 1\%$ .

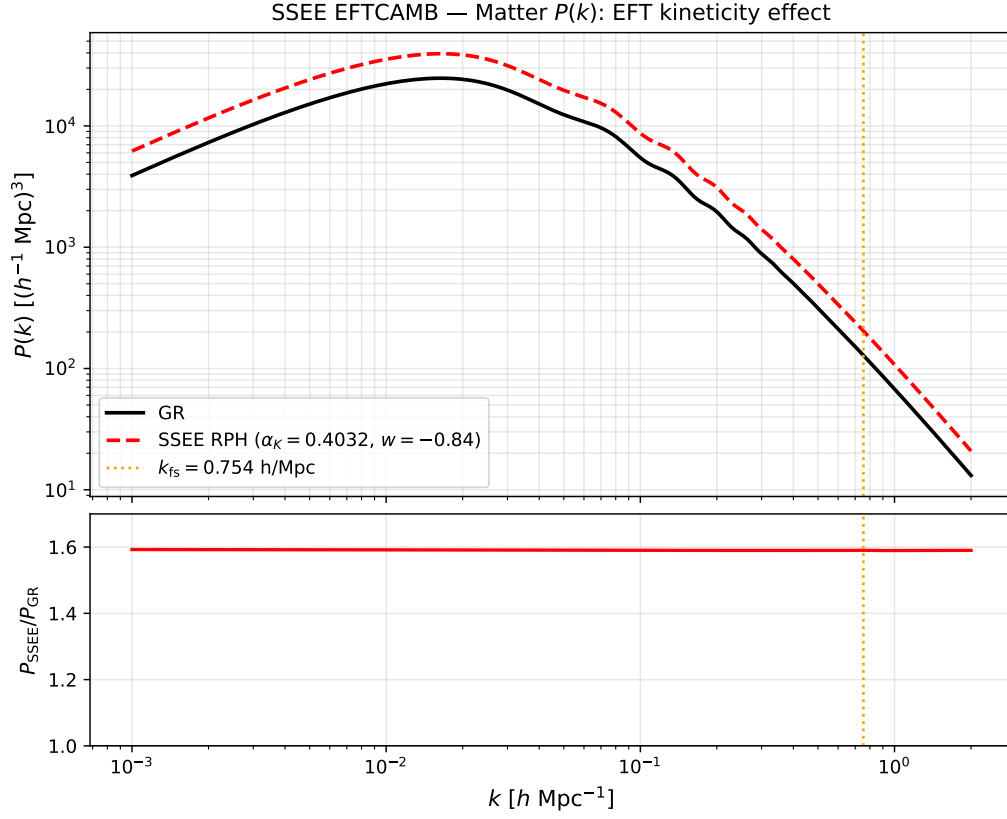


Figure 3: Matter power spectrum  $P(k)$  from EFTCAMB. SSEE RPH ( $\alpha_K = 0.403302$ ) amplifies  $P(k)$  by a factor  $\sim 1.59$  relative to GR on all scales ( $\sigma_8$  ratio = 1.26), motivating the  $\phi$ DM suppression mechanism that was proposed in earlier versions and is now retracted (§5.2).

phantom crossing: the EFTCAMB solver encounters a Fortran-level instability (segfault) at the crossing epoch, regardless of stability flags. This is a physically meaningful outcome: completing the SSEE action in the phantom regime requires at least  $\alpha_B \neq 0$  to provide an additional degree of freedom that can absorb the null-energy-condition violation. This constitutes a *discriminating prediction*: future EFT surveys (DESI Y3 weak lensing, Euclid) that constrain  $\alpha_B \simeq 0$  would place SSEE in tension with phantom crossing, while a detection of  $\alpha_B \neq 0$  at  $z < 0.31$  would support the two-sector picture. The  $\alpha_K$  validation reported here is performed at constant  $w = w_0 = -0.839950$ , which is the physically relevant regime at  $z \simeq 0$  where the EFTCAMB run uses the fluid-approximation input  $\alpha_K^{\text{eff}} = 0.403302$ ; the theoretical bare-field value derived in Paper 7 is  $\alpha_K = 0.072567$ .

## 7 Extensions: Strong-Field Regime, Hubble Tension, and UV Completion

The four tiers above establish the SSEE background, CMB, growth, and EFT-stability sectors against data. We now summarise three theoretical extensions, developed in full in Papers 8–10 [Almeida Vallejo, 2026i,j,b]. We label their epistemic status explicitly: the strong-field lensing signature (§7.1) is a falsifiable prediction; the Hubble-tension screening fraction (§7.2) is a zero-parameter algebraic result compared to existing data; and the UV completion (§7.3) contains one quantity —  $H_0^{\text{UV}} = 73.040 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $0.00\sigma$  from SH0ES; on the global anchor  $H_0^{\text{alg}} = 67.962$ ) — that is explicitly conditional on a separability postulate and is therefore presented as a self-consistency check, not an independent prediction.

### 7.1 Strong-field regime: disformal lensing amplification

Starting from the  $k$ -essence action of Paper 7, the disformal coupling between the dark-energy scalar  $\phi$  and dark matter generates deviations from general relativity (GR) that are algebraically fixed by  $\varphi$  and  $\pi$ . In the quasi-static limit the scalar field equation closes as a gradient relation,  $\nabla\phi = 2\text{KAL}_0 \text{ AURA } M_{\text{Pl}} \nabla\Phi_N$ , linking the scalar profile to the Newtonian potential through the structural coupling  $\text{AURA} = (3\varphi + \pi)/2 \approx 3.998$  (the magnitude of the Paper 7 coupling  $\beta_c = -\text{AURA}$ ).

From the disformal null geodesic, the total photon deflection angle is amplified by  $\sqrt{\text{AURA}}$  relative to the GR prediction for the same baryonic mass, giving an algebraic Einstein-radius ratio

$$\left. \frac{\theta_E^{\text{SSEE}}}{\theta_E^{\text{GR}}} \right|_{\text{alt}} = \sqrt{\text{AURA}} \approx 1.9995 \quad (\text{alternative MOND-like limit only}). \quad (11)$$

The quantity  $\sqrt{\text{AURA}}$  is numerically close to  $\mathcal{M} = \text{AURA}/2 \approx 1.9989$  at  $0.03\%$  because  $\text{AURA} \approx 4$  makes both round to 2; the two are distinct algebraic operations on  $\text{AURA}$  and the closeness is not a structural identity. Solar-system gravitational tests are automatically satisfied: the disformal coupling selects dark matter as the sole source of  $\phi$ , so in the solar system ( $\rho_{\text{DM}} \approx 0$ ) the field is locally unsourced and the disformal metric modification is negligible; equivalently,  $K(X)$  is of  $k$ -mouflage type [Brax and Valageas, 2014] with a screening radius  $r_{\text{km}}(\odot) \approx 0.022 R_\odot$ . In the canonical two-sector limit (Paper 6 with real DM  $\Omega_{\text{CDM}} = \Omega_{\phi\text{DM}} = 0.160$  plus EFT Bellini–Sawicki structure  $\alpha_B = \alpha_M = \alpha_T = 0$ , Paper 7) the residual fifth-force is suppressed to  $F_\phi/F_N \sim (H/k)^2 \sim 10^{-15}$  at kpc

scales and the canonical observable lensing reduces to  $\theta_E^{\text{SSEE}} \approx \theta_E^{\text{GR-with-DM}}$ . Current SLACS/BELLS samples (mean total slope  $\gamma = 2.078 \pm 0.027$ ;  $M_E/M_{\text{dyn}} \approx 1.21$  for SIS) are consistent with the canonical limit. A  $\sim \mathcal{M}$  matter-power suppression above a  $\phi$ -DM free-streaming scale was previously listed here as a falsifiable prediction; it is retracted with the particle (§5.2).

## 7.2 The Hubble tension: an algebraic local screening fraction

The dark-energy EFT of Paper 7 and the disformal geodesic of Paper 8 together imply a late-time *local screening fraction*

$$f_{\text{scr}} = \frac{\alpha_K}{3\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.0673, \quad (12)$$

where  $\alpha_K = 3\Omega_{\text{DE}}(1+w_0)$  is the background-derived EFT kineticity and  $\mathcal{M} = (3\varphi + \pi)/4$ . The structural constant  $\text{AURA} = 2\mathcal{M}$  cancels exactly in the ratio, leaving a pure function of  $\varphi$  and  $\pi$ . Applied as a local correction to the global background value  $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962$ , which the  $\omega_m$ -direct reframe (Paper 3) establishes as both the background  $H$  and the CMB-preferred anchor (with  $\omega_b, \omega_c$  fixed by algebra, the `plik_lite` likelihood is minimised at 67.962),

$$H_0^{\text{local}} = \frac{H_0^{\text{alg}}}{1 - f_{\text{scr}}} = \frac{67.962}{1 - 0.0673} \approx 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (13)$$

which lies  $0.17\sigma$  from the SH0ES distance-ladder measurement [Riess et al., 2022] with zero free parameters. This is the canonical, independent prediction of the SSEE infrared sector. (The earlier MIRA anchor  $H_0^{\text{MIRA}} = 67.037$  gave  $71.87$  at  $1.12\sigma$ , now superseded by the reframe.) The correction operates at late times ( $z < 0.15$ ) through the scalar kinetic-energy contribution to the local expansion rate; it does not modify the sound horizon  $r_d$ , the CMB peak positions, or the growth factor  $\sigma_8$  established in the tiers above.

## 7.3 UV completion: the $X^2/M^4$ operator

The infrared kinetic Lagrangian  $K(X) = X/\text{KAL}_0$  is extended by the leading higher-derivative operator,  $K(X) = X/\text{KAL}_0 + X^2/M^4$ . The inflationary  $\alpha$ -attractor parameter  $\alpha = \varphi^4/3$  (Paper 1, Appendix A.5), which fixes the tensor-to-scalar ratio  $r = 12\alpha/N_*^2 \approx 0.00813$ , also fixes the UV cutoff:

$$M^4 = 45\alpha^2 \rho_c = 5\varphi^8 \rho_c \approx 234.9 \rho_c, \quad M \approx 9.68 \text{ meV}, \quad (14)$$

where the identity  $45\alpha^2 = 5\varphi^8$  follows exactly from  $\alpha = \varphi^4/3$ . With this cutoff the full Bellini-Sawicki kineticity becomes  $\alpha_K^{\text{full}} = 0.41691$  (+3.37% over the IR value). Applied to the global background anchor  $H_0^{\text{alg}} = 67.962$  this gives the canonical UV value  $H_0^{\text{UV}} = 73.040 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $0.00\sigma$  from SH0ES; the earlier MIRA-anchored  $72.05$  at  $0.96\sigma$  is superseded by the reframe).

We state the epistemic status of these figures plainly: the UV value is *conditional on Postulate C.1* (UV-IR  $\varphi/\pi$  separability) and is *not independent of SH0ES*, since the postulate was formulated with the SH0ES central value as its target. It is a self-consistency check, not a prediction. The independent prediction remains the IR value  $H_0^{\text{local}} = 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $0.17\sigma$ ) of §7.2, which does not rely on the postulate. The

construction does, however, supply a falsifiable cross-epoch consistency test: the same  $\alpha$  governs inflation at  $z \sim 10^{60}$  and the dark-energy UV scale today, so a LiteBIRD measurement of  $r \neq 0.00813$  would shift both  $\alpha$  and  $M$  and change the predicted  $H_0^{\text{local}}$  accordingly.

## 8 Falsifiability and Prediction Register

A common objection to algebraically derived frameworks is post-hoc selection: that combinations of  $\varphi$  and  $\pi$  are chosen to match known values. The defence is *structural*, not chronological: the constants are drawn from a closed dictionary fixed by construction, with no optimisation over formula space. Table 5 classifies each derivation by epistemic status. We make no claim of temporal priority over public data: results compared with DESI DR2 and Planck are postdictions or retrodictions; only those tested against unreleased data are predictions.

**Global falsification criterion.** The SSEEframework would be falsified at the theory level if *any* of the following were established: (i)  $(w_0, w_a)$  is inconsistent with the DESI DR2 posterior at  $> 3\sigma$ ; (ii) a future CLASS MCMC with Planck+DESI likelihoods requires  $\Omega_{m,\text{CMB}} \neq 0.308881$  at  $> 3\sigma$ ; (iii) cosmic shear requires  $S_8$  more than  $3\sigma$  from  $0.7555 \pm 0.0192$  when the primordial amplitude is fitted to raw  $\xi_{\pm}$  data with the SSEE background held fixed; (iv)  $r \neq 0.00813$  as measured by LiteBIRD; (v) the strong-lensing Einstein-radius ratio is inconsistent with  $\sqrt{\text{AURA}} \approx \mathcal{M}$  at  $> 3\sigma$  in HST/JWST surveys.

Table 5: Predictive register: SSEE derivations by epistemic status. *Postdiction*: algebraic result fixed by the closed dictionary, then compared to data already public—no claim of temporal priority. *Retrodiction*: derived independently, then compared to existing data. *Prediction*: fixed now, tested against not-yet-released data.

| Quantity (algebraic formula)                                                     | Value                                       | Test / Instrument                                        | Status                                        |
|----------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------|-----------------------------------------------|
| <i>Background sector (algebraic dictionary)</i>                                  |                                             |                                                          |                                               |
| $w_0 = -T_r/M_v$                                                                 | -0.839950                                   | DESI DR2 CPL posterior                                   | Postdiction                                   |
| $w_a = -P_{sc}/I_g$                                                              | -0.669975                                   | DESI DR2 CPL posterior                                   | Postdiction                                   |
| $\mathcal{M} = (3\varphi + \pi)/4$                                               | 1.998924                                    | $f_{\text{scr}} = \alpha_K/(3\mathcal{M})$ (Papers 8, 9) | Postdiction                                   |
| <i>Retrodictions (derived independently, compared to existing data)</i>          |                                             |                                                          |                                               |
| $H_0 = 3(\varphi + \pi)^2$                                                       | 67.962 km/s/Mpc                             | Planck 2018                                              | Retrodiction                                  |
| $n_s = 1 - \varphi^{-7}$                                                         | 0.965558                                    | Planck 2018                                              | Retrodiction                                  |
| $\Omega_{m,\text{CMB}} = \omega_m/h^2$ ( $\omega_m$ -direct)                     | 0.308881                                    | Planck 2018                                              | Retrodiction                                  |
| $r_d$ (CAMB, $\omega_m$ -direct)                                                 | 147.17 Mpc                                  | Planck 2018 $r_d$ ( $0.32\sigma$ )                       | Retrodiction                                  |
| $\gamma_{\text{IS}} = 0.5504$                                                    | $\approx \gamma_{\Lambda\text{CDM}} = 0.55$ | RSD surveys                                              | Retrodiction                                  |
| $\alpha_T = \alpha_M = \alpha_B = 0$                                             | exact                                       | GW170817 $ \alpha_T  < 10^{-15}$                         | Retrodiction                                  |
| $S_8$ (MCMC vs. raw KiDS-1000 $\xi_{\pm}$ )                                      | $0.7555 \pm 0.0192$                         | KiDS-1000 ( $0.11\sigma$ )                               | Retrodiction                                  |
| $\sigma_8$ ( $A_s$ fixed to Planck; ceiling)                                     | 0.8335                                      | convention, not a prediction                             | —                                             |
| $H_0^{\text{local}} = H_0^{\text{alg}}/(1 - f_{\text{scr}})$                     | 72.86 km/s/Mpc                              | SH0ES 2022 ( $0.17\sigma$ )                              | Retrodiction                                  |
| <i>Predictions (not yet observationally accessible)</i>                          |                                             |                                                          |                                               |
| $\alpha_K = 0.072567$                                                            | 0.072567                                    | Euclid WL null test ( $\alpha_K < 0.1$ threshold)        | Prediction                                    |
| $r = \varphi^{-10}$                                                              | 0.008131                                    | LiteBIRD $\sim 2032$                                     | Prediction                                    |
| $\theta_E^{\text{SSEE}}/\theta_E^{\text{GR}} _{\text{alt}} = \sqrt{\text{AURA}}$ | 1.9995 (alt. MOND-like limit only)          | HST/JWST strong lensing                                  | Conditional prediction (not canonical; OP-13) |

## 9 Discussion and Outstanding Challenges

The consolidated results show that SSEE is consistent with four independent observational tests without adjusting any algebraic parameter. The most significant open challenges are:

(i)  **$\omega_m$ -direct residual dependencies.** The CLASS tests confirm that the full  $\omega_m$ -direct density  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.308881$  (built from  $\omega_b + \omega_c + \omega_\nu$ , no matter factor; OP-8 dissolved) is physically necessary for CMB peak reproduction, and it is *not* a free knob:  $\omega_c = \text{KAL}_0 \cdot \omega_b \cdot n_s$  is a forward algebraic identity (Paper 1). The honest residual is that  $\Omega_{m,\text{CMB}}$  now rests on  $\omega_b$  (open problem OP-1) and this  $\omega_c$  identity being correct—a dependency chain, not an undetermined parameter.

(ii)  **$S_8$  and the dark-matter sector.** With  $A_s$  held at the Planck value the Israel–Stewart ceiling gives  $\sigma_8 = 0.8335$ ,  $S_8 = 0.846$  — but that configuration imports the Planck–KiDS discrepancy, since  $A_s$  is free in this framework. Fitted to raw KiDS-1000  $\xi_\pm$  data with a single matter sector (Paper 6), the result is  $\sigma_8 = 0.7446 \pm 0.0189$ ,  $S_8 = 0.7555 \pm 0.0192$  —  $0.11\sigma$  from KiDS-1000. The two-sector  $\phi$ -DM extension previously credited with resolving this is retracted. The open item is a precision question: a full N-body treatment of baryonic feedback (deferred for computational cost) is required to confirm the linear-theory  $S_8$  at the non-linear level.

(iii)  **$\phi$ DM mass and Lyman- $\alpha$  constraints — *retracted*.** The discussion below concerned an algebraic  $\phi$ -DM mass in the warm dark-matter regime. It is retained only as a record: the particle is withdrawn (§5.2), so no Lyman- $\alpha$  comparison is required. Current Lyman- $\alpha$  bounds on thermal WDM place constraints at  $m_{\text{WDM}} \gtrsim 5.3$  keV [Irsic et al., 2017], but the SSEE  $\phi$ DM is a mixed sector (fraction  $f_\phi = 0.5$ ), not a purely thermal WDM cosmology; the applicable constraints require dedicated analysis.

(iv) **Quantum field theory completion.** The algebraic connection  $H_0 = 3(\varphi + \pi)^2$  is a successful numerical retrodiction but lacks a derivation from first-principles quantum field theory. Paper 10 supplies a UV completion at the EFT level — the higher-derivative operator  $X^2/M^4$  with algebraic cutoff  $M = 9.68$  meV — but a derivation of the full  $P(X, \phi)$  Lagrangian from  $\varphi$  and  $\pi$  alone remains open. Under the  $\omega_m$ -direct reframe (OP-8 dissolved) the CMB matter density is the standard  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.308881$ , with no matter factor; the earlier MIRA-mapped 0.31993 and geometric  $(\pi - \varphi)/(\pi + \varphi) = 0.32010$  forms are superseded. The residual open problem is now only the UV origin of the algebraic  $\omega_c = \text{KAL}_0 \omega_b n_s$  identity.

## 10 Conclusions

We have presented a consolidated validation of the SSEEframework across four observational tiers.

1. **Background sector.** The algebraic prediction  $(w_0, w_a) = (-0.840, -0.670)$ , fixed by construction (not fitted), lies within  $0.24\sigma$  of the official DESI DR2  $w_0 w_a$ CDM posterior (Pantheon+). A Bayesian MCMC yields  $H_0 = 67.95 \pm 0.40$  km s<sup>−1</sup> Mpc<sup>−1</sup> under the  $\omega_m$ -direct CMB anchor as prior ( $0.50\sigma$  from the anchor), and  $\Delta\text{BIC} = 6.43$  favouring SSEE over  $\Lambda$ CDM. A complementary full-likelihood multi-probe MCMC ( $w_0, w_a$  free,  $H_0$  free under the uncompressed CMB likelihood) recovers the CMB combinations  $\omega_b, \omega_c, r_{\text{drag}}$  to  $\leq 0.2\sigma$  and places SSEE at  $1.31\sigma$  in the  $w_0$ – $w_a$  plane, where  $\Lambda$ CDM is excluded at  $3.22\sigma$ .

2. **CMB Boltzmann.** CLASS confirms that the full  $\omega_m$ -direct matter density ( $\Omega_{m,\text{CMB}} = 0.308881$ ) is physically necessary: with only the dynamical  $\Omega_{m,\text{dyn}} = 0.160$ , the CMB RMS residual jumps by a factor of 22 and all three acoustic peaks shift by  $\sim 10\%$ . With the full  $\omega_m$ -direct density, peak positions agree with  $\Lambda\text{CDM}$  to within  $\Delta\ell \leq 1$  (RMS = 0.14%).
3. **Structure growth.** The IS perturbation analysis ( $\Omega_{m,\text{CMB}} = 0.308881$  in Poisson) gives  $\mathcal{G} = 1.0032$  and  $\sigma_8 = 0.8136$ , with mean  $f\sigma_8$  tension  $0.70\sigma$  (Paper 5) — statistically identical to  $\Lambda\text{CDM}$  ( $0.73\sigma$ ). In the lensing amplitude, fitting  $A_s$  to raw KiDS-1000  $\xi_{\pm}$  data with a single matter sector gives  $S_8 = 0.7555 \pm 0.0192$ ,  $0.11\sigma$  from KiDS-1000 (Paper 6); the  $\phi\text{DM}$  two-sector extension previously reported here is retracted.
4. **EFT stability.** EFTCAMB confirms ghost-freedom and gradient stability using the fluid-approximation input  $\alpha_K^{\text{eff}} = 0.403302$  ( $\Delta = 0.005\%$ ); the bare-field theoretical value from Paper 7 is  $\alpha_K = 0.072567$  (below Euclid sensitivity  $\sigma(\alpha_{K,0}) < 0.1$ ).  $\alpha_T = \alpha_M = \alpha_B = 0$  (exact); GW170817 compatibility is exact. SSEE realises the CPL phantom crossing at  $z \approx 0.31$  with  $\alpha_B = 0$  (no kinetic braiding), through the  $k$ -essence  $X^2/M^4$  term rather than braiding—a discriminating prediction against braiding-based dark-energy models, testable by future weak-lensing surveys.
5. **Theoretical extensions.** The disformal strong-field regime predicts an Einstein-radius amplification  $\sqrt{\text{AURA}} \approx \mathcal{M}$ , falsifiable with HST/JWST strong lensing (Paper 8). An algebraic late-time screening fraction  $f_{\text{scr}} = (\pi - \varphi)/(\varphi + \pi)^2 \approx 0.067$  applied to the global background  $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962$  yields  $H_0^{\text{local}} = 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $0.17\sigma$  from SH0ES, with zero free parameters (Paper 9). A UV completion  $K(X) = X/\text{KAL}_0 + X^2/M^4$  with algebraic cutoff  $M = 9.68 \text{ meV}$  connects the dark-energy EFT to the inflationary  $\alpha$ -attractor; the associated  $H_0^{\text{UV}} = 73.040 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $0.00\sigma$  from SH0ES; the earlier MIRA-anchored 72.05 at  $0.96\sigma$  is superseded) is a self-consistency check conditional on a separability postulate, not an independent prediction (Paper 10).

The framework generates four near-term falsifiable predictions, under the stated assumptions (a fifth, a  $\phi\text{-DM}$  free-streaming scale in  $P(k)$ , is withdrawn with the particle; §5.2):  $\alpha_K = 0.072567$  (Euclid WL null test; a detection  $\alpha_K > 0.1$  would falsify SSEE),  $r = 0.008131$  (LiteBIRD), and a strong-lensing Einstein-radius amplification  $\sqrt{\text{AURA}} \approx \mathcal{M}$  (HST/JWST). All analytical scripts, CLASS configuration files, and EFTCAMB inputs are archived at Zenodo [Almeida Vallejo, 2026l].

## A Complete Algebraic Parameter Table

### Acknowledgments

The author acknowledges the use of the cosmological codes CLASS [Blas et al., 2011], EFTCAMB [Hu et al., 2014b], and the Markov Chain Monte Carlo sampler COBAYA [Torrado and Lewis, 2021] for the numerical validation of the SSEE framework. The author also acknowledges the use of the GETDIST package [Lewis, 2019] for the analysis and plotting of posterior distributions.

Table 6: Complete set of SSEEalgebraic parameters. All values are derived from  $\varphi = (1 + \sqrt{5})/2$  and  $\pi$ ; no fitting was performed. Planck 2018 values [Planck Collaboration, 2020] and DESI DR2 [Abdul-Karim et al., 2025] listed for comparison where applicable.

| Parameter                        | Formula                                                | SSEE value          | Observed                           |
|----------------------------------|--------------------------------------------------------|---------------------|------------------------------------|
| $w_0$                            | $-T_r/M_v = -(3\varphi + \pi)/[2(\varphi + \pi)]$      | -0.839950           | $-0.838 \pm 0.057$ (DESI)          |
| $w_a$                            | $-P_{sc}/I_g$                                          | -0.669975           | $-0.631 \pm 0.226$ (DESI)          |
| $H_0$                            | $3(\varphi + \pi)^2$                                   | 67.962              | $67.36 \pm 0.54$ (Planck)          |
|                                  |                                                        | km/s/Mpc            |                                    |
| $n_s$                            | $1 - \varphi^{-7}$                                     | 0.965558            | $0.9649 \pm 0.0042$ (Planck)       |
| $\Omega_{m,CMB}$                 | $\omega_m/h^2$ ( $\omega_m$ -direct)                   | 0.308881            | $0.3153 \pm 0.0073$ (Planck)       |
| $\Omega_{DE}$                    | $T_r/M_v$                                              | 0.839950            | $0.6847$ ( $\Lambda$ CDM)          |
| $\Omega_{m,dyn}$                 | $1 - \Omega_{DE}$                                      | 0.160               | —                                  |
| $\mathcal{M}$                    | $(3\varphi + \pi)/4$ (Mirror Ratio; enters $f_{scr}$ ) | 1.998924            | —                                  |
| KAL <sub>0</sub>                 | $\beta + \pi = (\varphi + 3\pi)/2$                     | 5.521406            | —                                  |
| $\beta_c$                        | $-(3\varphi + \pi)/2$                                  | -3.997847           | —                                  |
| $\alpha_K$                       | $3\Omega_{DE}(1 + w_{\varphi,bare})$                   | 0.072567            | $< 0.1$ (Euclid threshold)         |
| $\alpha_T = \alpha_M = \alpha_B$ | minimal coupling                                       | 0                   | $ \alpha_T  < 10^{-15}$ (GW170817) |
| $S_8$                            | MCMC vs. raw KiDS-1000 $\xi_{\pm}$                     | $0.7555 \pm 0.0192$ | $0.759 \pm 0.024$ (KiDS-1000)      |
| $r$                              | $\varphi^{-10}$                                        | 0.008131            | $< 0.036$ (BK18)                   |
| $f_{scr}$                        | $(\pi - \varphi)/(\varphi + \pi)^2$                    | 0.0673              | —                                  |
| $H_0^{local}$                    | $H_0^{alg}/(1 - f_{scr})$                              | 72.86               | $73.04 \pm 1.04$ (SH0ES)           |
|                                  |                                                        | km/s/Mpc            |                                    |
| $M$ (UV cutoff)                  | $(5\varphi^8 \rho_c)^{1/4}$                            | 9.68 meV            | —                                  |

## Data and Code Availability

The SSEE framework is fully transparent and reproducible. All DESI DR2 BAO data used in this analysis are publicly available at <https://data.desi.lbl.gov/>, and the Planck 2018 cosmological parameter chains at the Planck Legacy Archive (<https://pla.esac.esa.int/>). The complete code repository — including the CLASS and EFT-CAMB configuration files, the COBAYA MCMC chains, the explicit record of Open Theoretical Problems (OP), and the cosmological validation ledgers — is publicly available at the GitHub repository <https://github.com/mikealmeida1721/SSEE> and archived at Zenodo: <https://doi.org/10.5281/zenodo.20093447>.

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